HR ANALYTICS — EXECUTIVE DASHBOARD

CodeAlpha Internship Project Report

Employee Attrition • Satisfaction • Performance • Workforce & Hiring Trends

Project	CodeAlpha HR Analytics Dashboard
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Primary Tool	Microsoft Power BI
Repository	ManyaSain/CodeAlpha_HR_Analytics_Dashboard
Repository Status	Public GitHub repository
Report Date	27 August 2026

This report documents the completed HR Analytics dashboard project. The GitHub repository contains the Power BI report, supporting employee and dimension data files, dashboard preview, README and .gitignore. The dashboard is designed to turn workforce data into interactive HR insights and support data-driven decision-making.

1. Project Overview

The HR Analytics — Executive Dashboard is an interactive Microsoft Power BI solution for analyzing employee attrition, job satisfaction, management/performance ratings, workforce composition and hiring trends. Power BI forecasting is also used to provide an indication of future workforce or hiring requirements.

2. Problem Statement

HR teams often have employee information available but need a clearer way to identify attrition patterns, compare departments and job roles, understand employee experience indicators, and monitor workforce trends. This project addresses that need through a single interactive analytical dashboard.

3. Objectives

- Track employee and workforce-related metrics.
- Analyze employee attrition and turnover patterns.
- Analyze job satisfaction and management/performance ratings.
- Identify attrition patterns across departments and job roles.
- Analyze gender-wise attrition distribution.
- Analyze historical hiring/workforce trends.
- Use Power BI forecasting to estimate future trends.
- Provide an interactive dashboard for HR decision-making.

4. Dataset & Data Model

The GitHub repository currently contains the main employee dataset and supporting dimension/reference files. The repository lists Employee.csv, EducationLevel.csv, PerformanceRating.csv, RatingLevel.csv, SatisfiedLevel.csv and DimDate.txt, along with the Power BI report and dashboard image. This structure supports a more organized Power BI data model rather than keeping all analytical information in one flat table.

Repository Data File	Role in Project
Employee.csv	Primary employee/workforce data used for HR analysis.
EducationLevel.csv	Education-level reference information.
PerformanceRating.csv	Performance/management rating-related data.
RatingLevel.csv	Rating-level reference information.
SatisfiedLevel.csv	Job-satisfaction level reference information.
DimDate.txt	Date dimension used for time-based analysis and trends.

5. Data Preparation

- Data cleaning and validation were performed before visualization.
- Required categorical and numerical fields were organized for analysis.
- Power Query was used within Power BI for transformation and preparation.
- DAX measures were created for analytical calculations.
- Data was prepared for interactive visualization and forecasting.

6. Key Performance Indicators

The repository README documents the following major KPIs in the dashboard:

KPI	Purpose
Total Employees	Overall workforce size.
Active Employees	Employees currently active.
Attrition Count	Number of employees associated with attrition.
Attrition Rate	Attrition as a proportion of the workforce.
Average Age	Average employee age.
Average Salary	Average employee salary.

7. Dashboard Visualizations

7.1 Attrition Count by Department and Job Role

Compares attrition across departments and job roles to identify areas with higher employee exits.

7.2 Employee Job Satisfaction Distribution

Shows employee distribution across satisfaction ratings and supports an attrition-based comparison.

7.3 Manager Rating vs Attrition

Examines the relationship between management ratings and employee attrition.

7.4 Hiring Trend & Forecast

Displays historical workforce/hiring trends and uses Power BI forecasting to estimate future trends.

7.5 Attrition Distribution by Gender

Provides a gender-wise view of employee attrition and supports workforce comparison.

8. Interactive Slicers

The repository README specifies interactive filters for Department, Job Role, Gender and time-related analysis. These slicers allow HR users to move from an overall workforce view to specific employee segments.

9. Technology Stack

Technology	Application
Microsoft Power BI	Dashboard development, data modeling and visualization.
Power Query	Data cleaning and transformation.
DAX	Measures and analytical calculations.
Microsoft Excel	Data preparation/source-data handling.
Data Visualization	Presentation of HR metrics and patterns.
Forecasting Analytics	Future hiring/workforce trend estimation.

10. Forecasting Approach

The Hiring Trend & Forecast visual uses Power BI's forecasting functionality on historical workforce/hiring trends. The forecast is intended as an analytical estimate that can support workforce planning; it should not be interpreted as a guaranteed future outcome.

11. Business Value & Insights Supported

- Identify departments and job roles with comparatively higher attrition.
- Compare employee satisfaction patterns with attrition.
- Examine management/performance rating patterns associated with attrition.
- Compare gender-wise attrition distribution.
- Monitor workforce/hiring trends over time.
- Support short-term workforce and hiring planning.

12. GitHub Repository Review

The public repository was reviewed directly. It currently contains 10 commits and the following visible project assets: .gitignore, CodeAlpha_HR_Analytics.pbix, DimDate.txt, EducationLevel.csv, Employee.csv, HR_Analytics_Dashboard.png, PerformanceRating.csv, README.md, RatingLevel.csv and SatisfiedLevel.csv. The README describes the project overview, objectives, KPIs, visuals, slicers, tools, data preparation, forecasting and repository structure.

Current repository structure observed:

Asset	Current purpose/status
CodeAlpha_HR_Analytics.pbix	Main Power BI project file.
Employee.csv	Primary employee dataset.
EducationLevel.csv	Supporting dimension/reference data.
PerformanceRating.csv	Supporting performance-rating data.
RatingLevel.csv	Supporting rating-level data.
SatisfiedLevel.csv	Supporting satisfaction-level data.
DimDate.txt	Date dimension/supporting time data.
HR_Analytics_Dashboard.png	Dashboard preview image.
README.md	Project documentation and repository overview.
.gitignore	Git ignore configuration.

13. Recommended Final Repository Structure

For a polished portfolio submission, the current root-level files can remain functional, but the repository can be made more professional by organizing them into folders as follows:

```
CodeAlpha_HR_Analytics_Dashboard/
├── README.md
├── .gitignore
├── PowerBI/
│ └── CodeAlpha_HR_Analytics.pbix
├── Dataset/
│ ├── Employee.csv
│ ├── EducationLevel.csv
│ ├── PerformanceRating.csv
│ ├── RatingLevel.csv
│ ├── SatisfiedLevel.csv
│ └── DimDate.txt
├── Dashboard/
│ └── HR_Analytics_Dashboard.png
└── Documentation/
├── Project_Report.pdf
└── Data_Dictionary.xlsx
```

This organization is a recommendation for presentation quality; the existing files already provide the core project assets.

14. Limitations

- Dashboard conclusions depend on the quality and completeness of the underlying employee data.
- Observed relationships between satisfaction, ratings and attrition should not automatically be treated as causal.
- Forecast values are estimates based on historical patterns.
- Additional factors such as compensation, tenure, workload, management style and career growth may also affect attrition.

15. Future Scope

- Add machine-learning based employee attrition prediction.
- Create an employee attrition-risk scoring model.
- Include additional HR variables such as tenure, overtime, promotion, training and compensation.
- Add automated data refresh and scheduled reporting.
- Create drill-through pages for detailed department and job-role analysis.

16. Conclusion

The CodeAlpha HR Analytics Dashboard demonstrates practical use of Power BI, Power Query and DAX to transform employee data into an interactive workforce-analysis solution. The project combines attrition, satisfaction, performance/management ratings, gender analysis and hiring trends with forecasting. Its GitHub repository contains the Power BI report, supporting datasets, dashboard preview, README and .gitignore, making it suitable as a structured internship and portfolio project.

17. Project Deliverables

Deliverable	Purpose
Power BI (.pbix)	Interactive HR Analytics dashboard.
Employee & supporting CSV/TXT files	Source and dimensional data used by the report.
Dashboard PNG	Visual preview for portfolio/GitHub.
README.md	Project documentation.
.gitignore	Repository housekeeping.
Project Report (this PDF)	Detailed project documentation.
Data Dictionary (recommended next)	Field-by-field explanation of all dataset columns.

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